

Introduction to NLP

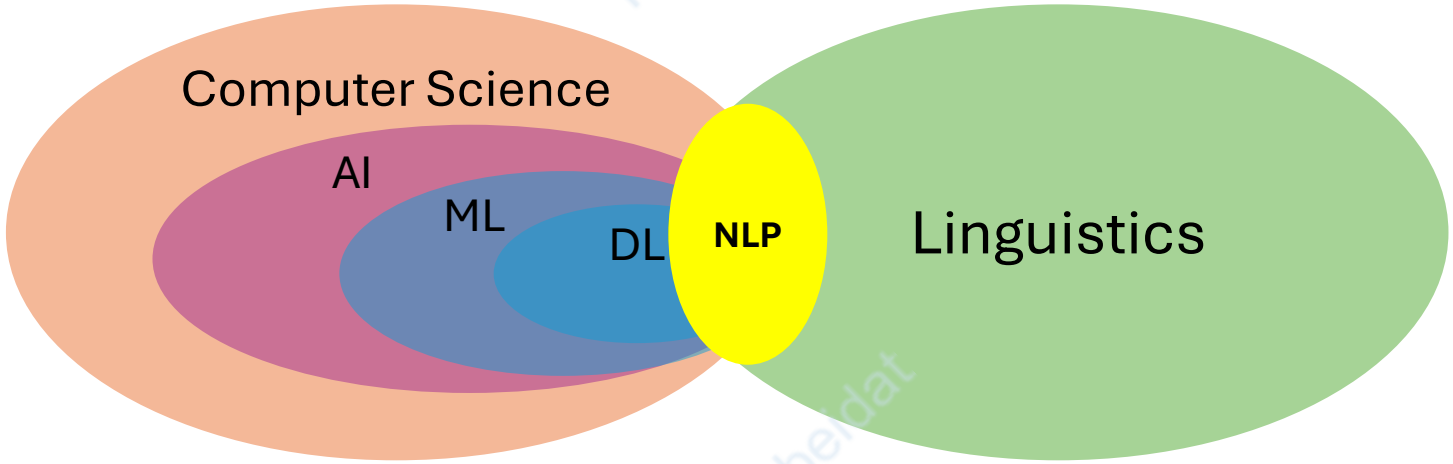
Rasha Obeidat, PhD

AI445

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What is NLP?

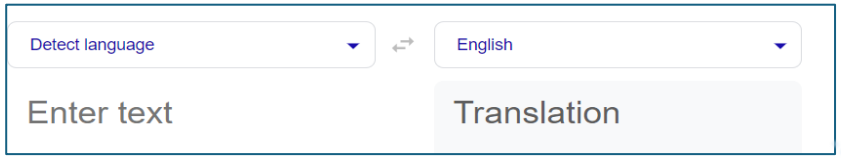
- **Definition:** Natural Language Processing (**NLP**) is the ability of a computer program to understand human language as it's spoken and written.
- **Goals of NLP**:
  - To enable machines to read, interpret, and generate human language.
  - To facilitate human-computer interaction.



Applications of NLP

- Virtual assistants (e.g., Siri, Alexa)
- Machine translation (e.g., Google Translate)
- Sentiment analysis (e.g., on social media)
- Text summarization (e.g., news or article summarizers
- Chatbots (e.g., customer service bots)



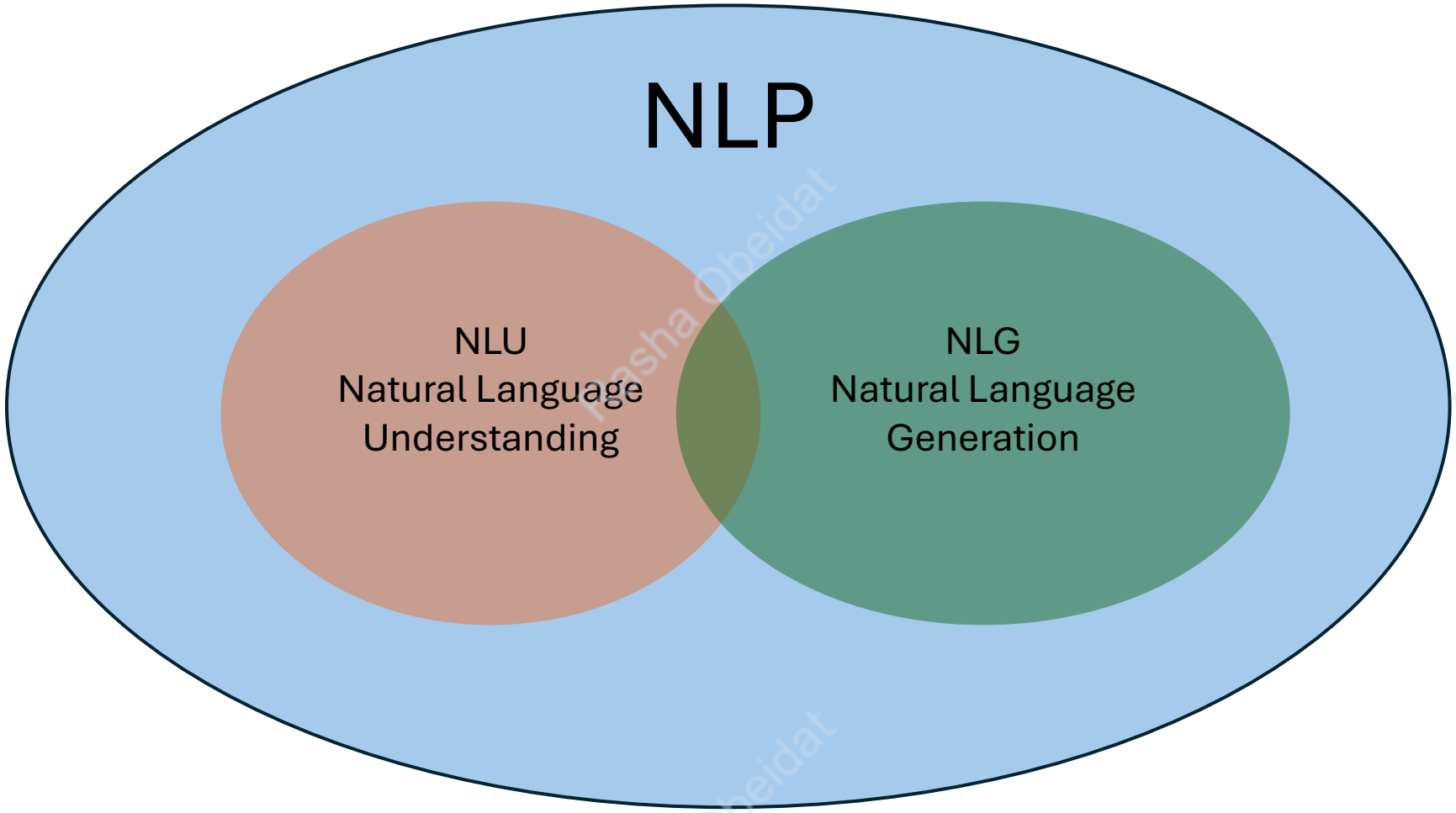








Components of NLP



Natural Language Understanding (NLU)

- **Definition**: NLU is the branch of NLP that focuses on interpreting and comprehending human language. It enables machines to understand input in natural language, including grammar, context, and meaning.
- **Key Tasks**:
  - **Named Entity Recognition (NER)**: Identifying and classifying entities (e.g., people, locations) in text.
  - **Sentiment Analysis**: Determining the sentiment or emotional tone (positive, negative, neutral) of text.
  - **Text Classification**: Categorizing text into predefined labels or topics (e.g., spam detection, news categorization).
  - **Machine Reading Comprehension (MRC)**: Automatically answering questions about a given text or passage by "reading" and understanding it

Natural Language Generation (NLG)

- **Definition**: NLG is the process of automatically generating coherent and contextually appropriate text or speech in human language based on structured data or input.
- **Key Tasks**:
  - **Text Summarization**: Condensing a long piece of text into a shorter version while retaining key information.
  - **Machine Translation**: Translating text from one language to another (e.g., English to Spanish).
  - **Dialogue Generation**: Generating conversational responses in chatbots or virtual assistants (e.g., generating coherent replies to user queries).
  - **Report Generation**: Creating structured reports or documents from raw data (e.g., financial or medical reports).

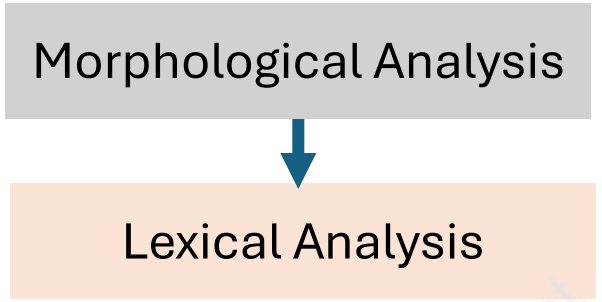
Levels of Understanding in NLP

Morphological Analysis

- **Morphology**: Study of word forms and morphemes (e.g., root words, prefixes, suffixes) and their relationship to the meaning of the word .
  - **Focus**: internal word structure.
  - **Typical lexical analysis tasks:**
    - Lemmatization, Stemming, Root extraction (very important in Arabic NLP)
    - Morphological tagging (gender, number, case, tense…)
    - Diacritization (Arabic — strongly morphological)

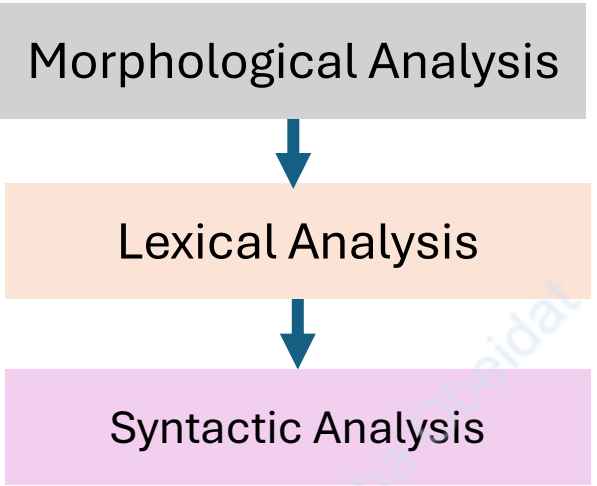
Levels of Understanding in NLP

- **Morphology**
- **Lexical Structure:** Study of words as units of language, including their form, identity, grammatical category (e.g., noun, verb), and basic meanings as entries in the lexicon.
  - **Focus:** representation and properties of individual words.
  - **Typical lexical analysis tasks:**
    - Part-of-Speech (POS) tagging
    - Named Entity Recognition (NER)
    - Multiword expression detection
    - Lexicon construction and lookup



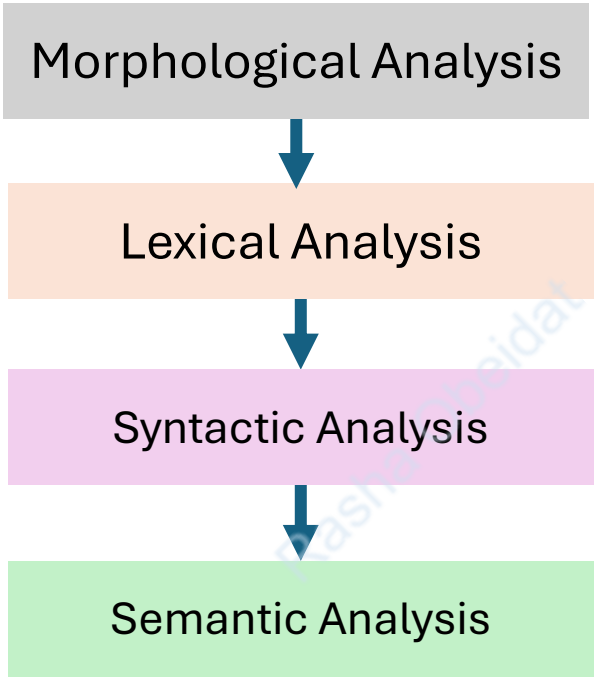
Levels of Understanding in NLP

- **Morphology**
- **Lexical structure**
- **Syntax:** Study of sentence structure and the rules that govern how words combine to form grammatical sentences (e.g., phrase structure, word order, agreement).
  - **Focus:** Structure of sentences and relationships between words in a sentence.
  - **Typical syntactic analysis tasks:**
    - Constituency parsing
    - Dependency parsing
    - Chunking (shallow parsing)
    - Grammar checking
    - Agreement detection



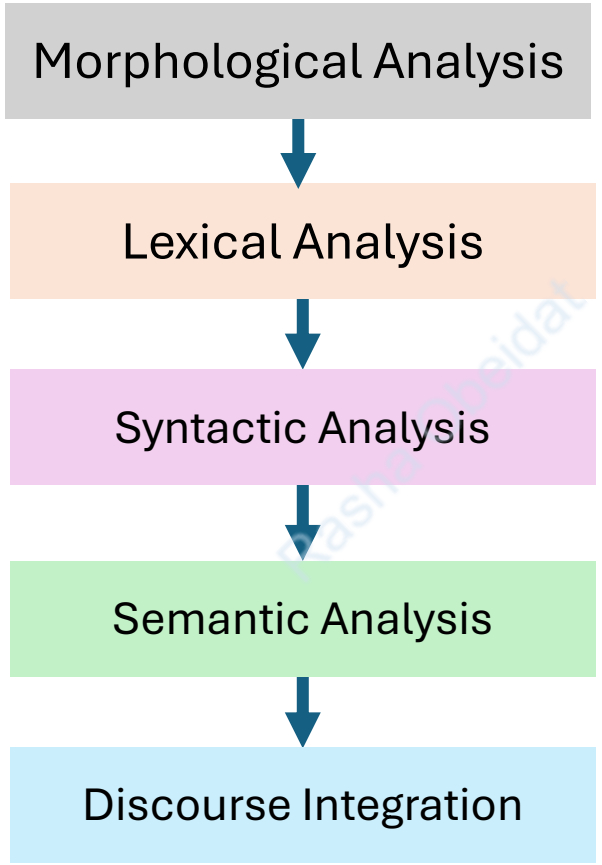
Levels of Understanding in NLP

- **Morphology**
- **Lexical structure**
- **Syntax**
- **Semantics:** Study of meaning at the sentence level, describing how words and phrases combine to express coherent meaning in sentences and larger text units (e.g., paragraphs).
  - **Focus:** interpretation of meaning and relationships between concepts in context.
  - **Typical semantic analysis tasks:**
    - Semantic Role Labeling (SRL)
    - Semantic parsing
    - Natural Language Inference (NLI) / textual entailment
    - Relation extraction



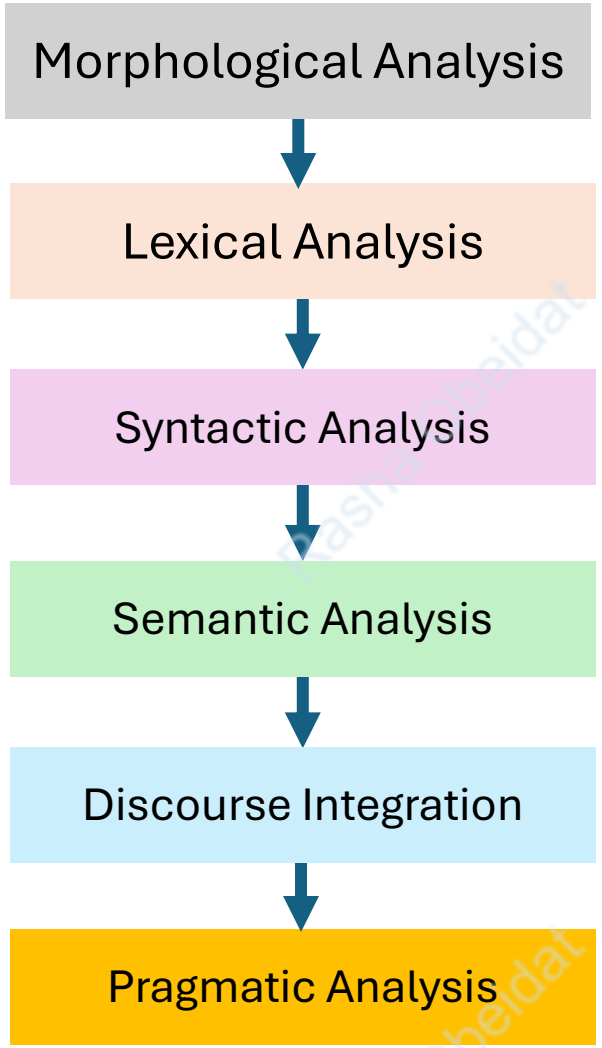
Levels of Understanding in NLP

- **Morphology**
- **Lexical structure**
- **Syntax**
- **Semantics**
- **Discourse Integration:** Study of how meaning is connected across multiple sentences, focusing on identifying the larger context in which smaller language units (e.g., words, phrases, or sentences) are interpreted.
  - **Focus:** coherence, context, and relationships across sentences and larger text units.
  - **Typical discourse analysis tasks:**
    - Coreference resolution
    - Topic segmentation
    - Text summarization
    - Machine Reading Comprehension (mainly ***semantic*** level with strong discourse support)



Levels of Understanding in NLP

- **Morphology**
- **Lexical structure**
- **Syntax**
- **Semantics**
- **Discourse Integration**
- **Pragmatics:** Study of how context influences the interpretation of meaning in communication, focusing on how language is used in real-life situations beyond literal meaning.
  - **Focus:** speaker intention, context, and language use in real-world interaction.
  - **Typical pragmatic analysis tasks:**
    - Dialogue systems and conversational agents
    - Intent detection
    - Sentiment and emotion analysis (context-aware)
    - Sarcasm and irony detection



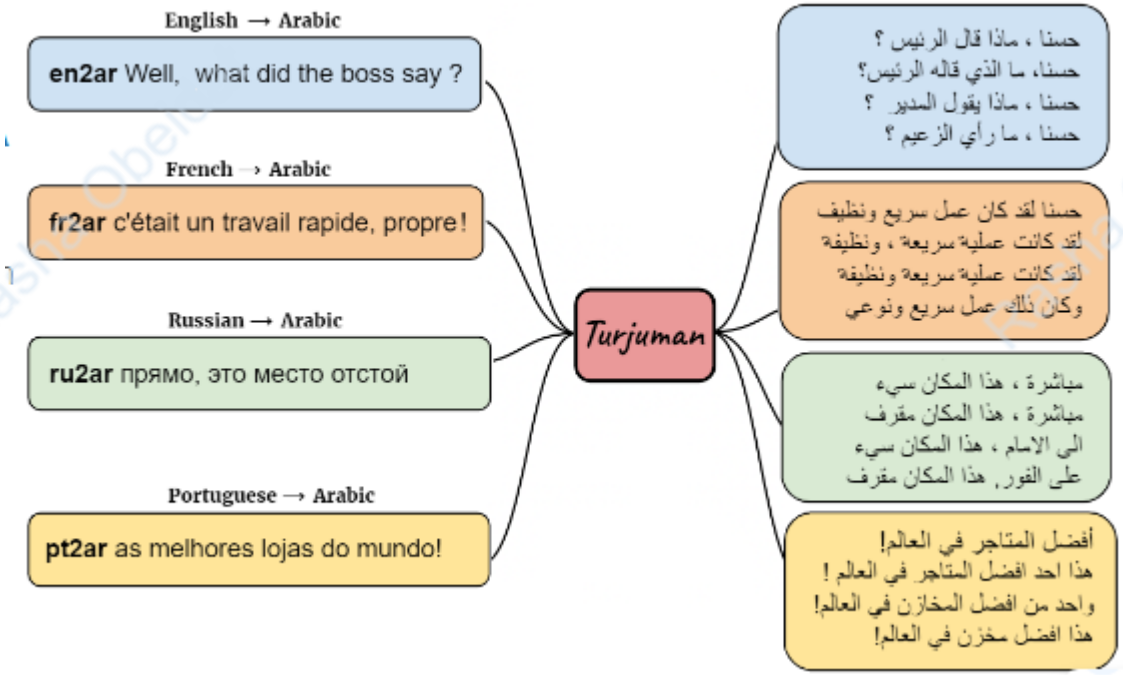
A General Framework for NLP Systems

- Formally, create a function to map an input X (language) into an output Y. Examples:

Task: Translation

Input X: Text

Output Y: another text



https://demos.dlnlp.ai/turjuman/

A General Framework for NLP Systems

- Formally, create a function to map an input X (language) into an output Y. Examples:

  Task: Machine Reading Comprehension

  Input X: Text

  Output Y: Response text

الفقرة: ولد أبو بكر الصدِّيق في مكة سنة 573م بعد عام الفيل بسنتين وستة أشهر، وكان من أغنياء قُريش في الجاهليَّة، فلما دعاه النبي مُحمد إلى الإسلام أسلمَ دون تردد، فكان أول من أسلم مِن الرجال الأحرار. ثم هاجر أبو بكر مُرافقًا للنبي مُحمد من مكة إلى المدينة، وشَهِد غزوة بدر والمشاهد كلها مع النبي مُحمد، ولما مرض النبي مرضه الذي مات فيه أمر أبا بكر أن يَؤمَّ الناس في الصلاة.

السؤال: أين ولد أبو بكر الصديق رضي الله عنه؟

اجابة جيدة: في مكة

اجابة سيئة: ولد أبو بكر الصدِّيق في مكة

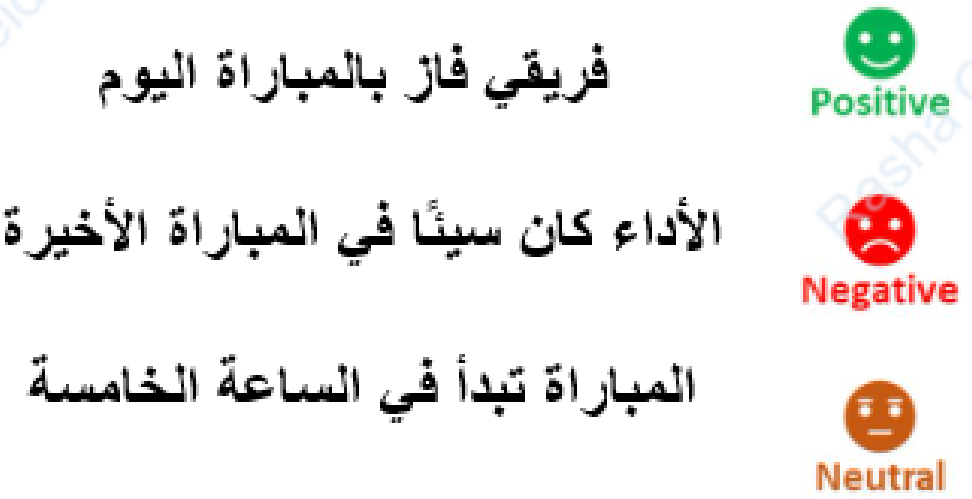
A General Framework for NLP Systems

- Formally, create a function to map an input X (language) into an output Y. Examples:

Task: Sentiment Analysis

Input X: Text

Output Y: label



Common Difficulties in NLP Tasks

1. **Ambiguity:** refers to the phenomenon where a word, phrase, or sentence can have multiple interpretations, making it challenging for machines to determine the intended meaning without contextual understanding.
   - **Lexical Ambiguity**: Words with multiple meanings.
     Example: bank → river bank vs. financial institution.
   - **Syntactic Ambiguity**: Sentences that can be parsed in more than one way.
     Example: I saw the man with the telescope.
   - **Semantic Ambiguity**: Uncertainty in meaning even after syntax is resolved.
     Example: The chicken is ready to eat.
   - **Pragmatic Ambiguity**: Meaning depends on context, speaker intent, or social norms.
     Example: Can you open the window? (request vs. ability)

Example: Pragmatic Ambiguity



1



2



3

Example: Pragmatic Ambiguity



4



5



6

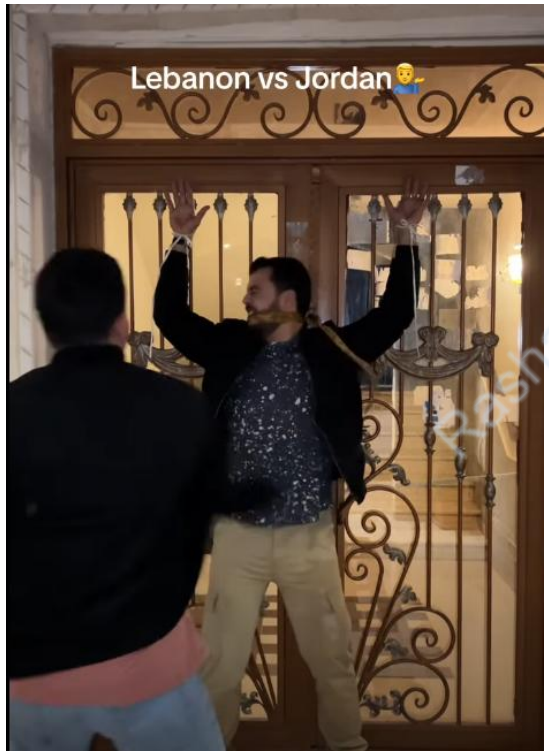
Example: Pragmatic Ambiguity



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Example: Pragmatic Ambiguity









Common Difficulties in NLP Tasks

2. **Low-Frequency Words:** Words that appear infrequently in the training data are often poorly represented, leading to challenges in understanding or predicting their meaning.
   - **Example**: Rare scientific or technical terms, or terms from classical languages (e.g., Classical Arabic).

قعقعت الأبواب في الرياح العاتية

Common Difficulties in NLP Tasks

3. Language Variability and Creativity

- Multiple ways to express the same idea.
- Use of idioms, metaphors, sarcasm, slang, or regional dialects (e.g., Jordanian Arabic vs. MSA).
- Informal or noisy text (e.g., social media) makes processing harder.









Common Difficulties in NLP Tasks

4. Low-Resource Languages

- Many languages have limited available data (e.g., text corpora, annotations), making it difficult to build effective NLP models.
- Example cases (real-world type):
  - Some Amazigh (Berber) varieties
  - Indigenous languages in South America
  - Minority dialects in remote villages



- **NLP problem:**
  - Almost **no training data**
  - No dictionary or grammar resources

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Common Difficulties in NLP Tasks

5. Conjugation and Morphological Variations. Languages with rich morphology (e.g., Arabic, Finnish) pose challenges due to complex word forms resulting from conjugation, pluralization, and gender variations.

- **Example**: A single root word can have many different forms, making it hard for models to handle all variants.
  - The word يحب can take many forms:
    - أحب
    - تحب
    - تحبين
    - يحبون
    - etc.
- If the model doesn't account for these conjugated forms, it might misinterpret the sentiment or fail to recognize the emotional tone of a sentence.

Common Difficulties in NLP Tasks

6. Metaphors and Idiomatic Expressions

- Figurative language, such as metaphors, idioms, and non-literal expressions, are difficult for models to interpret accurately.
- **Example**: in sentiment analysis:

ما شاء الله، وصلت على الموعد كالعادة!



- **Example**: in machine translation

حماتك بتحبك

Literal meaning: Your mother-in-law loves you

idiomatic meaning (the correct): You arrived at just the perfect time; we're about to eat!

Common Difficulties in NLP Tasks

7. Long-Range Dependencies: occur when related words or meanings appear far apart in a sentence or across sentences, requiring NLP models to connect distant information to understand meaning correctly.

- **Common NLP tasks affected:**
  - Coreference resolution
  - Machine translation
  - Question answering
  - Parsing and semantic understanding
- **Example 1:** *"The book that John gave to Sarah, which she read over the weekend, was **fascinating**."*
  - Who/What is the fantastic? John, Sara, the book?
- **Example** 2: زار خالد صديقه في عمّان بعد أن أنهى امتحاناته، وكان سعيدًا جدًا بذلك.
  - Who was happy? خالد or صديقه؟
- Long-range dependencies are difficult because models must remember earlier information while processing later words.

Common Difficulties in NLP Tasks

6. **Negation:** can significantly change the meaning of a sentence, and models often struggle to properly account for it.
   - **Example**: handling negation is crucial for sentiment analysis if the negation is not properly handled.



Key Milestones in NLP

1950: Turing Test (Alan Turing)

- Proposed a test to evaluate machine intelligence through conversation.

1960s-1980s:Rule-based NLP

- **E.g., ELIZA (Joseph Weizenbaum)** First chatbot, demonstrating simple conversational abilities with pattern matching.

1980s-2010: Statistical NLP

- Shift to probabilistic models like **Hidden Markov Models** (HMMs), enabling scalable language processing.
- NLP relied on manual feature design (e.g., **bag-of-words**, **n-grams**) with algorithms like **SVMs** and **Naive Bayes**.

Key datasets (e.g., **Penn Treebank**, **CoNLL-2003**) became benchmarks for model evaluation.

2013-2015: Sequence Modeling & RNNs & non-contextualized word representations

- **RNNs** and **LSTMs** improved tasks like machine translation with **Seq2Seq** models, enabling better handling of long-range dependencies. Introduction of **Word2Vec** and **GloVe**, which represented words as static vectors in a fixed **2015-2017: Attention Mechanism**

2015-2017: Attention Mechanism

- The **attention mechanism** (Bahdanau et al.) improved translation by focusing on relevant parts of input sequences, boosting Neural Machine Translation (NMT).

2017: The Transformer Model

- **Transformers** replaced RNNs with **self-attention**, allowing efficient parallel processing and becoming the foundation for models like **BERT** and **GPT**.

2018-Present: Pre-trained Language Models

- Models like **BERT** and **GPT-3** set new benchmarks with large-scale pre-training, revolutionizing NLP tasks such as text generation and translation.

Main Tasks in Natural Language Processing

- **Core Language Processing**
  - Tokenization and text normalization
  - Morphological analysis (stems, lemmas, affixes)
  - Part-of-speech tagging
  - Named entity recognition
  - Text Classification
  - Syntactic parsing (dependency and constituency)
  - Semantic role labeling
- **Text Understanding**
  - Word sense disambiguation
  - Coreference resolution
  - Sentiment analysis
  - Text classification (topic, intent, stance, etc.)
  - Natural language inference (NLI)
  - Question answering

Main Tasks in Natural Language Processing

- **Text Generation**
  - Machine translation
  - Text summarization (extractive and abstractive)
  - Dialogue systems and chatbots
  - Text simplification and paraphrasing
  - Automatic speech recognition (speech → text)
  - Text-to-speech (text → speech)
- **Linguistic Resources and Representation**
  - Corpus creation and annotation
  - Language modeling (n-gram, neural, transformer-based)
  - Word and sentence embeddings
  - Multilingual and cross-lingual modeling

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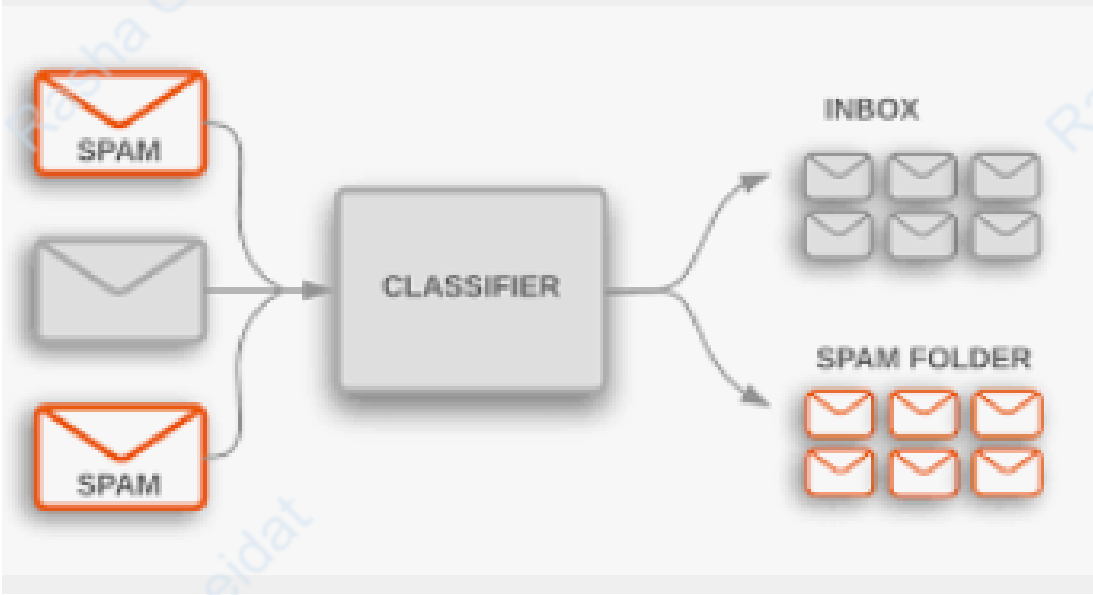
Main Tasks in Natural Language Processing

- **Information Extraction and Retrieval**
  - Keyword and relation extraction
  - Event extraction
  - Document and passage retrieval
  - Knowledge base population

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Natural Language Processing Tasks

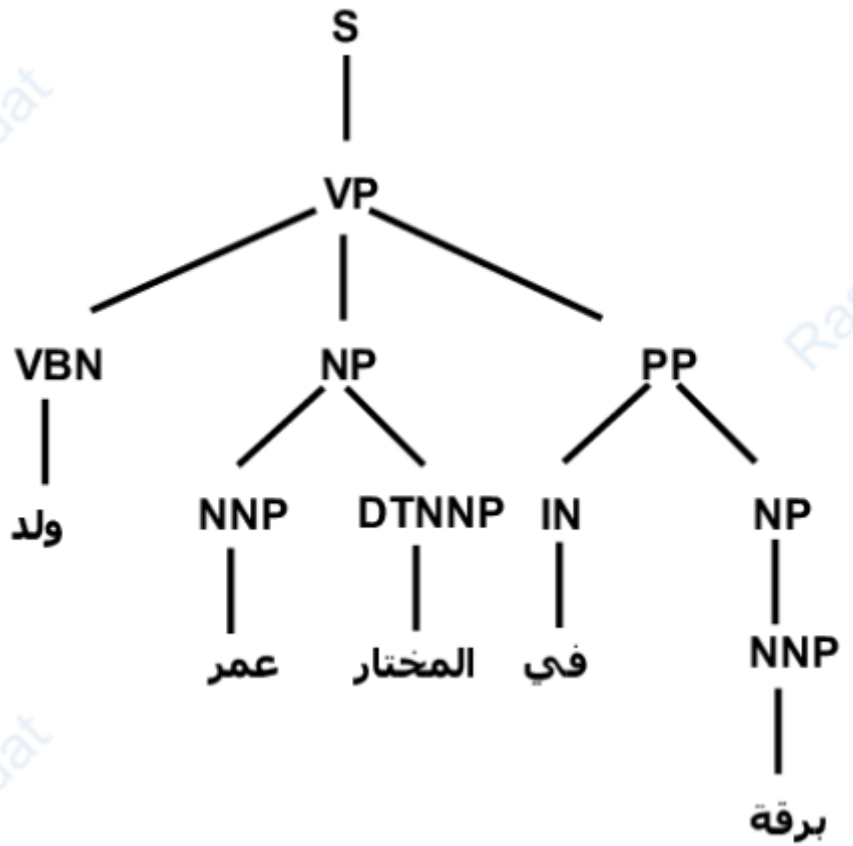
- **Text Classification**
- Assigning categories or labels to text.
- Examples: Sentiment analysis, spam detection, topic classification.



Natural Language Processing Tasks

- Formally, create a function to map an input X (language) into an output Y. Examples:

Task: Parsing



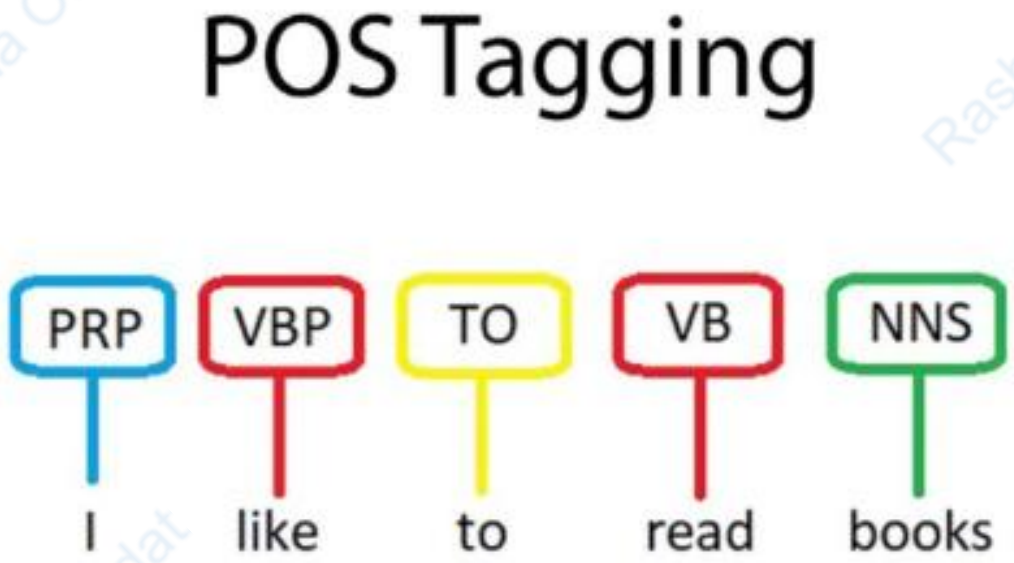
Natural Language Processing Tasks

- **Named Entity Recognition (NER)**
  - Identifying and classifying entities (e.g., persons, locations, organizations) in text.
  - Example: Extracting company names from news articles.



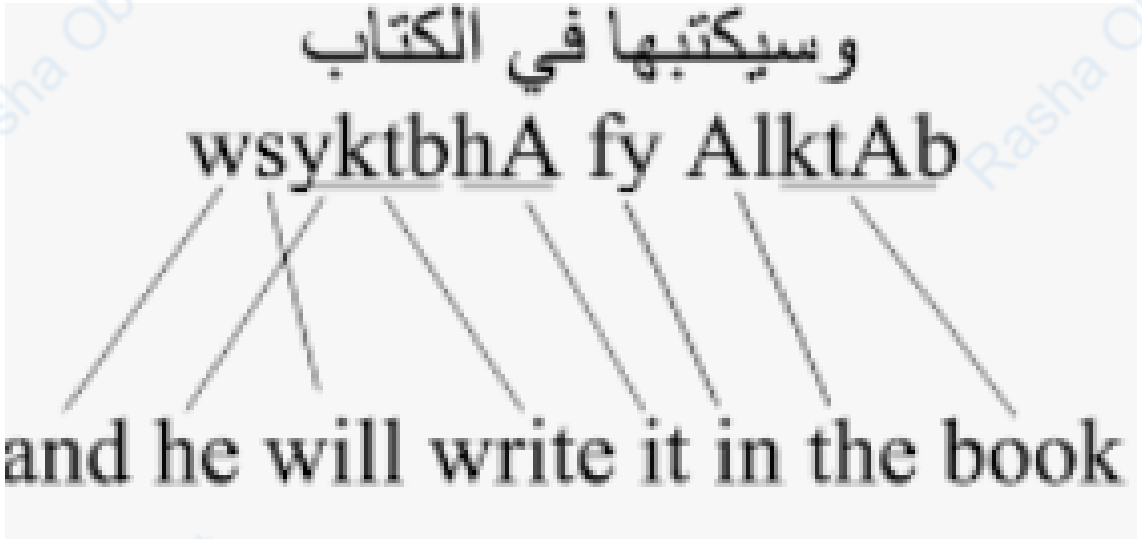
Natural Language Processing Tasks

- **Part-of-Speech (POS) Tagging**
- Assigning grammatical tags (e.g., noun, verb, adjective) to words in a sentence.



Natural Language Processing Tasks

- **Machine Translation (MT):** automatically translating text from one language to another.
- Example: Google Translate.



Natural Language Processing (NLP) Tasks

- **Question Answering (QA):** extracting precise answers from text given a user's query.
- **Example**: Answering factual questions from Wikipedia articles.
- How it differs from MRC? Has a retrieval components !

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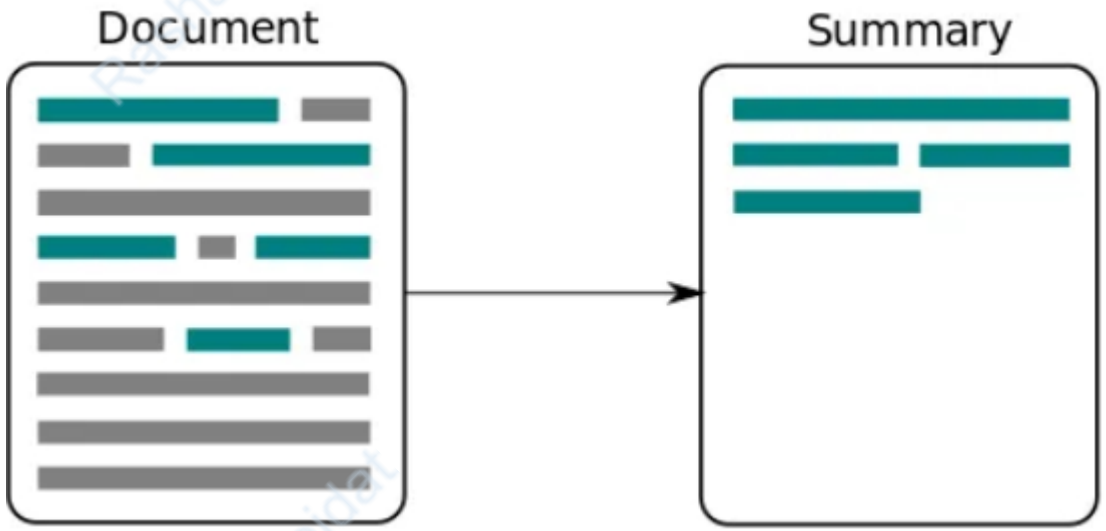
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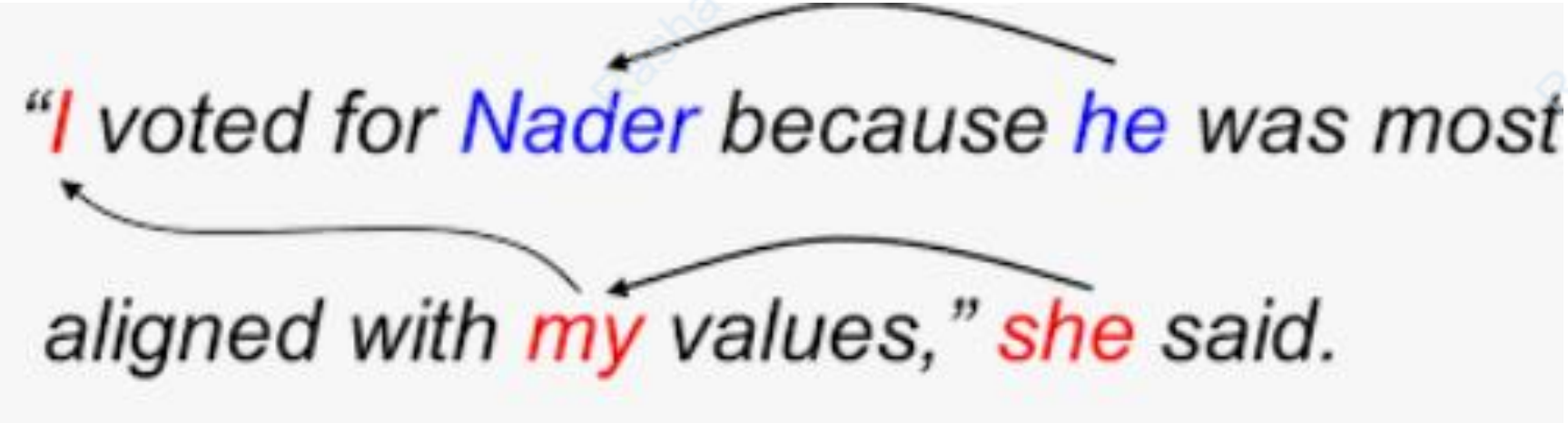
Natural Language Processing Tasks

- **Text Summarization:** Generating concise summaries of longer documents.
- **Example**: Summarizing news articles or research papers.



Natural Language Processing Tasks

- **Coreference Resolution:** Identifying which words refer to the same entity.
- **Example**: Resolving pronouns in a paragraph.





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